STAT 253: Probability & Statistics
UNIT II - Random Variables

April 3, 2023

Unit Objectives

This unit introduces students to concepts and computational techniques associated with random variables. After studying this unit students should be able to;

1. Define a random variable and distinguish between discrete and continuous random variables
2. Define, describe and recognise probability mass functions (pmf) and probability density functions (pdf)
3. Compute the probabilities of the various numerical values a random variable assumes.
4. Compute numerical characteristics of a random variable

Reference

Montgomery, D. and Runger, G. (2014). Applied Statistics and Probability for Engineers, Sixth Edition. John Wiley and Sons, Inc.

Random Variable

We shall not always be interested in a random experiment but rather in some result of its outcome.

A **random variable** is a function that assigns a real number to each outcome (event) in a sample space of a random experiment. Mathematically;

$$X : \Omega \to \mathbb{R}$$

Example: A fair coin is tossed twice

$$\Omega = \{HH, HT, TH, TT\}$$

Let X be the number of heads, so that

$$X(HH) = 2, \;\; X(HT) = X(TH) = 1, \;\; X(TT) = 0$$

II-1: Discrete Random Variable

A random variable is said to be discrete if takes on finite or countably infinite set of values in $\mathbb{R}$

Examples

1. The number of customers awaiting to be served at a bank
2. Number of scratches on a surface
3. Proportion of defective parts among 1000 tested
4. Number of transmitted bits received in error

II-1-1: Probability Mass Function

The probability distribution of a random variable X is a description of the probabilities associated with the possible values of X

For a discrete random variable X with possible values $x_1, x_2, \cdots, x_k$ a probability mass function (pmf) is a function such that

1. $f(x_i) \geq 0$
2. $\sum_{i=1}^{k} f(x_i) = 1$
3. $f(x_i) = \mathbb{P}(X = x_i)$; $f(x)$ is the probability that $X = x$

Example 1

Construct a probability mass function for the number of heads in throwing a coin thrice

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$X = \{3, 2, 1, 0\}$$

X	0	1	2	3
P(X = x)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Verify properties (1) and (2) on page 6.

Example 2

Once a pmf of X is presented, it is possible to calculate all types of events in the sample space. $\mathbb{P}(X \leq a)$, $\mathbb{P}(X < a)$, $\mathbb{P}(X \geq a)$,$\mathbb{P}(X > a)$,$\mathbb{P}(a < X < b)$, $\mathbb{P}(a \leq X \leq b)$ $\mathbb{P}(a < X \leq b)$.

Compute the ff using Example 1

$$\mathbb{P}(X \leq 1) =$$
$$\mathbb{P}(X < 1) =$$
$$\mathbb{P}(X > 2) =$$
$$\mathbb{P}(X \leq 3) - \mathbb{P}(X \leq 1) =$$
$$\mathbb{P}(1 < X < 3) =$$
$$\mathbb{P}(1 \leq X \leq 3) =$$
$$\mathbb{P}(1 < X \leq 3) =$$

Exercise

1 Determine whether each of the following can be a probability mass function of a discrete random variable

$$(a)\ \ f(x) = \frac{1}{2}(x-2); x = 1, 2, 3, 4$$

$$(b)\ \ g(x) = \frac{1}{10}(x+1); x = 0, 1, 2, 3$$

2 A discrete random variable is given by

$$f(x) = c(x+1), x = 0, 1, 2, 3$$

(a) Find the value of the constant c

(b) Find $\mathbb{P}(0 < X < 2)$ and $\mathbb{P}(X > 1)$

Distribution Function

When dealing with a random variable, X for constants a,b ($b < a$). We are mostly interested in one or several of the probabilities defined on page 8. For this reason we calculate $\mathbb{P}(X \leq x)$ which tells us almost everything about X.

Definition

Let X be a random variable, the function

$$F : (-\infty, +\infty) \to \mathbb{R}^+$$
$$x \to F(x) = \mathbb{P}(X \leq x)$$

$$F(x) = \mathbb{P}(X \leq x) = \sum_{x_i \leq x} f(x_i), \quad x \in (-\infty, +\infty)$$

$F(x)$ represents the probability that the random variable, X takes on a value that is less than or equal to x.

Properties of the CDF

(a) F(x) is non decreasing function of x (i.e if $x \leq y$, then $F(x) \leq F(y)$)

(b) $\lim_{x \to \infty} F(x) = F(\infty) = 1$

(c) $\lim_{x \to -\infty} F(x) = F(-\infty) = 0$

(d) $\mathbb{P}(x_1 < X \leq x_2) = F(x_2) - F(x_1)$

(e) For all $x \in \mathbb{R}$ we have $\mathbb{P}[X > x] = 1 - F(x)$

Example

In the experiment of flipping a fair coin twice, let X be the number of tails and calculate F(x), the distribution function of X, and then sketch its graph.

$$X = \{0, 1, 2\}$$

Its pmf is

X	0	1	2
$\mathbb{P}(X = x)$	0.25	0.5	0.25

$\mathbb{P}(X \leq x) = 0$; if $x < 0$

For $0 \leq x < 1$, we have

$$F(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X = 0) = \mathbb{P}(\{HH\}) = \frac{1}{4}$$

Example Cont'd

For $1 \leq x < 2$, we have

$$F(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X = 0 \text{ or } X = 1) = \mathbb{P}(\{HH\}, \{HT\}, \{TH\})$$
$$= \frac{3}{4}$$

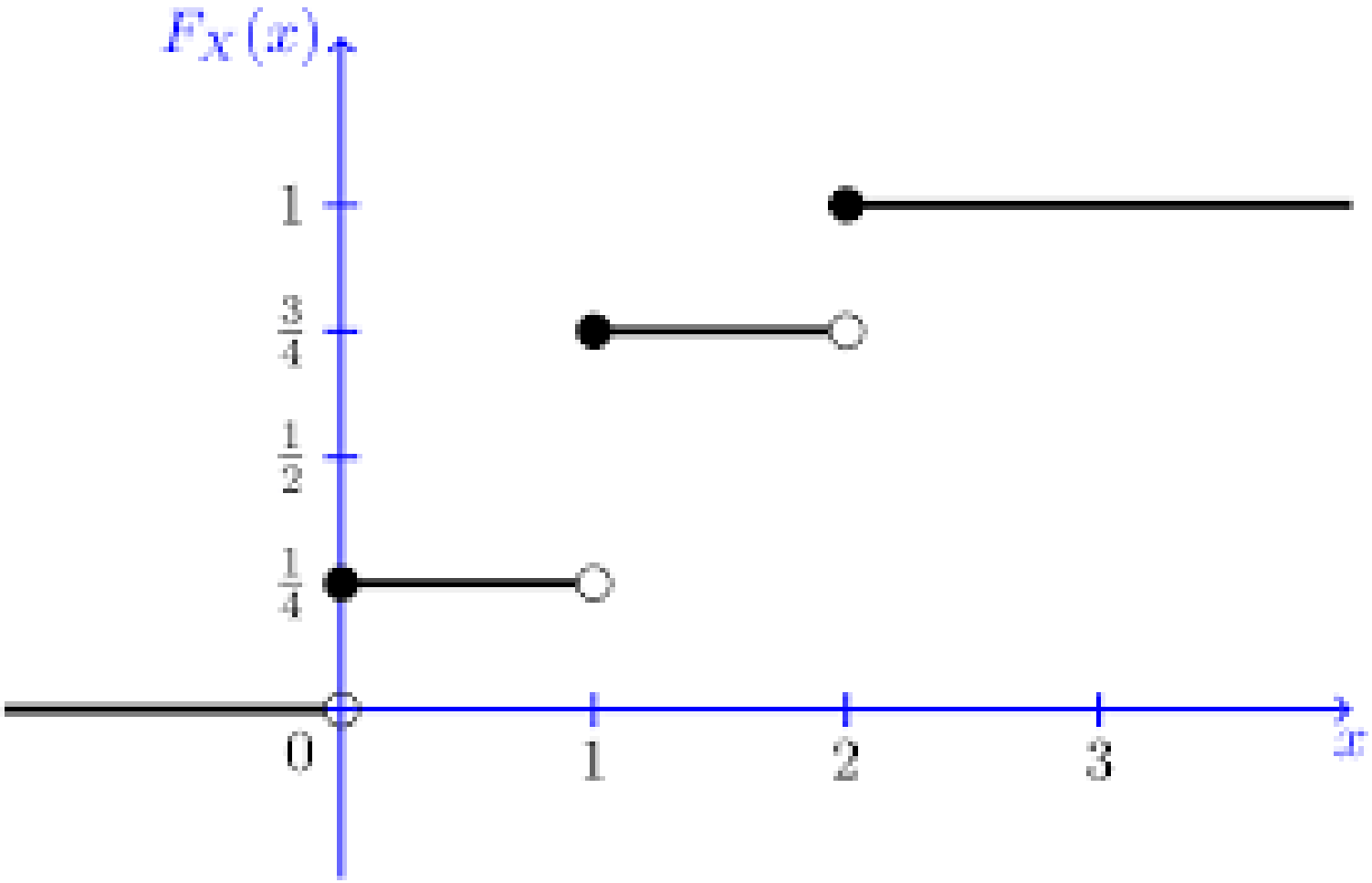
For $x \geq 2$, we have

$$F(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X = 0 \text{ or } X = 1 \text{ or } X = 2)$$
$$= \mathbb{P}(\{HH\}, \{HT\}, \{TH\}, \{TT\}) = 1$$

Hence we have;

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{4} & 0 \leq x < 1 \\ \frac{3}{4} & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

Graph



Exercise

Consider the following probability distribution

X	1	2	3	4
$\mathbb{P}(X = x)$	0.25	0.5	0.125	0.125

1. Find $F(2)$, $F(3)$, $F(3)$
2. Find the formula for $F(x)$ and sketch its graph

Distribution Function

Theorem

If the range of a discrete random variable X consists of the values $x_1 < x_2 < \cdots < x_n$ then $p(x_1) = F(x_1)$ and

$$p(x_i) = F(x_i) - F(x_{i-1}), i = 2, 3, \ldots, n$$

Distribution Function

Example(Discrete Case)

If the distribution function of X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{16} & 0 \leq x < 1 \\ \frac{5}{16} & 1 \leq x < 2 \\ \frac{11}{16} & 2 \leq x < 3 \\ \frac{15}{16} & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

Find the probability mass function (pmf).

Distribution Function

Solution

$$p(0) = \frac{1}{16}$$

$$p(1) = F(1) - F(0) = \frac{5}{16} - \frac{1}{16} = \frac{4}{16} = \frac{1}{4}$$

$$p(2) = F(2) - F(1) = \frac{11}{16} - \frac{5}{16} = \frac{6}{16} = \frac{3}{8}$$

$$p(3) = F(3) - F(2) = \frac{15}{16} - \frac{11}{16} = \frac{4}{16} = \frac{4}{16} = \frac{1}{4}$$

$$p(4) = F(4) - F(3) = 1 - \frac{15}{16} = \frac{1}{16}$$

CDF TABLE

Event concerning X	Probability of the event in terms of F	Event concerning X	Probability of the event in terms of F
$X \leq a$	$F(a)$	$a < X \leq b$	$F(b) - F(a)$
$X > a$	$1 - F(a)$	$a < X < b$	$F(b-) - F(a)$
$X < a$	$F(a-)$	$a \leq X \leq b$	$F(b) - F(a-)$
$X \geq a$	$1 - F(a-)$	$a \leq X < b$	$F(b-) - F(a-)$
$X = a$	$F(a) - F(a-)$		

Exercise

The distribution function of a random variable X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{4} & 0 \leq x < 1 \\ \frac{1}{2} & 1 \leq x < 2 \\ \frac{1}{12}x + \frac{1}{2} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

Compute the following quantities $\mathbb{P}(X < 2)$, $\mathbb{P}(X = 2)$, $\mathbb{P}(1 \leq X < 3)$, $\mathbb{P}(X > \frac{3}{2})$, $\mathbb{P}(X = \frac{5}{2})$, $\mathbb{P}(2 < X \leq 7)$

The answers are $\frac{1}{2}, \frac{1}{6}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{3}$ respectively.

Expectation and Variance

Two numbers often used to summarize a probability distribution for a random variable X are the mean and the variance. The mean is a measure of the center or middle of the probability distribution, and the variance is a measure of the dispersion, or variability in the distribution.

Definition (Expectation)

The mean or expected value of the discrete random variable, X denoted as μ or $\mathbb{E}$, is

$$\mu = \mathbb{E}(X) = \sum_x xf(x)$$

Example

The number of email messages received per hour has the following distribution:

X = number of messages	10	11	12	13	14	15
$f(x)$	0.08	0.15	0.30	0.20	0.20	0.07

Determine the mean number of messages received per hour.

Interpretation: On the average we would expect ——– emails per hour.

Expectation

Let X be a discrete random variable, then for any function $g : \mathbb{R} \to \mathbb{R}$, $g(X)$ is also a discrete random variable and has expectation

$$\mathbb{E}(g(X)) = \sum_{\forall x} g(x)f(x) \tag{1}$$

Let a be a constant, then the following properties hold;

1. $$\mathbb{E}(c) = c$$
2. $$\mathbb{E}[cg(X)] = c\mathbb{E}[g(X)]$$

Expectation

For linear function $g(x) = ax + b$ where a, b are constants, we have

$$[3]\mathbb{E}[g(X)] = a\mathbb{E}(X) + b$$

We show proofs of [1], [2] and [3]. These rules are also applicable if X is continuous.

Example

Equation (1) is known as the **law of the unconscious statistician** It enables one to calculate the expected value of the random variable $g(X)$ without deriving its probability mass function.

Example: The probability mass function of a discrete random variable X is given by

$$f(x) = \begin{cases} \frac{x}{15} & x = 1, 2, 3, 4, 5 \\ 0 & \text{elsewhere} \end{cases}$$

What is the expected value of $X(6 - X)$?

$$\mathbb{E}(X(6-X)) = 5 \times \frac{1}{15} + 8 \times \frac{2}{15} + 9 \times \frac{3}{15} + 8 \times \frac{4}{15} + 5 \times \frac{5}{15} = 7$$

Variance

The population variance of X, denoted as σ^2 or $V(X)$, is

$$\begin{aligned}\sigma^2 = V(X) =& \mathbb{E}(X-\mu)^2 = \sum_x (x-\mu)^2 f(x) \\ &= \sum_x x^2 f(x) - \mu^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2\end{aligned}$$

The standard deviation of X is $\sigma = \sqrt{\sigma^2}$

The larger (smaller) σ^2 is, the more (less) spread in the possible values of X about the population $\mu = \mathbb{E}(X)$.

Variance

Let X be a discrete random variable and a, b are two real numbers then

$$\mathbb{V}(a) = 0$$
$$\mathbb{V}(aX) = a^2\mathbb{V}(X)$$
$$\mathbb{V}(aX + b) = a^2\mathbb{V}(X)$$

We can easily show proof!

Example

What is the variance of the random variable X, the outcome of rolling a fair die?

X	1	2	3	4	5	6
$f(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

What is the variance of X?

$$\mathbb{E}(X) = \sum_{x=1}^{6} xf(x) = \frac{1}{6}\sum_{x=1}^{6} = \frac{1}{6}(1+2+3+4+5+6) = \frac{7}{2}$$

$$\mathbb{E}(X^2) = \sum_{x=1}^{6} x^2 f(x) = \frac{1}{6}\sum_{x=1}^{6} = \frac{1}{6}(1+4+9+16+25+36) = \frac{91}{6}$$

$$V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} \quad (2)$$

Exercise

Suppose that, for a discrete random variable X, $\mathbb{E}(X) = 2$ and $\mathbb{E}(X(X-4)) = 5$ Find the variance and the standard deviation of $-4X + 12$. Ans $\mathbb{V}(-4X + 12) = 144$ and $\sigma_{-4X+12} = 12$

Bernoulli Distribution

A single trial of an experiment may result in one of the two mutually exclusive outcomes, such as, head or tail in a toss of a coin (If we are interested in heads, we may call it a success(s); tails is then a failure (f).), yes or no in an election, dead or alive as a person comes out of a surgery, male or female in a child birth. Such a trial is called a Bernoulli trial.

If a fuse is inspected, it is either "defective" or it is "good." So the experiment of inspecting fuses is a Bernoulli trial. A good fuse may be called a success, a defective fuse a failure.

Bernoulli Distribution

A random variable X is said to have a Bernoulli distribution if it assumes the values 0 and 1 for two outcomes, $X(s) = 1, X(f) = 0$. If p is the probability of a success, then $1 - p$ (sometimes denoted q) is the probability of a failure.

Definition (Bernoulli)

A random variable is called Bernoulli with parameter p if its probability mass function is given by equation (3)

$$P(X = x) = \begin{cases} p \ ; x = 1 \\ 1 - p \equiv q \ ; x = 0 \\ 0 \ \text{otherwise} \end{cases} \tag{3}$$

Expectation and Variance

The expected value and variance of a Bernoulli random variable X, with parameter p, is

$$\mathbb{E}(X) = 0 \times \mathbb{P}(X = 0) + 1 \times \mathbb{P}(X = 1) = p$$

$$\mathbb{E}(X^2) = 0 \times \mathbb{P}(X = 0) + 1 \times \mathbb{P}(X = 1) = p$$

$$\mathbb{V}(X) = p - p^2 = p(1 - p) = pq$$

Hence, for a Bernoulli random variable X with parameter, p

$$\mathbb{E}(X) = p, \;\; \mathbb{V}(X) = p(1 - p), \;\; \sigma = \sqrt{p(1 - p)} \qquad (4)$$

Example

If in a throw of a fair die the event of obtaining 4 or 6 is called a success, and the event of obtaining 1, 2, 3, or 5 is called a failure, then

$$X = \begin{cases} 1 \text{ if } 4 \text{ or } 6 \text{ is obtained} \\ 0 \text{ otherwise} \end{cases} \tag{5}$$

is a Bernoulli random variable with parameter $p = \frac{1}{3}$. Therefore its pmf is

$$\mathbb{P}(X = x) = \begin{cases} \frac{2}{3} & ; x = 0 \\ \frac{1}{3} & ; x = 1 \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

$\mathbb{E}(X) = p = \frac{1}{3}, \ \mathbb{V}(X) = p(1-p) = \frac{2}{9}$

Binomial Distribution

They are generalizations of Bernoulli trials. Some examples are tossing a coin n times and observing the number of successes, heads or tails, a sequence of n shots may result in a number of hits or misses.

If n Bernoulli trials all with probability of success p are performed independently, then X, the number of successes is called binomial with parameters n and p. The set of possible values of X is $\{0, 1, 2, \cdots, n\}$

Thus, a Bernoulli random variable is just a binomial random variable with parameters (1, p).

Conditions of a Binomial Distribution

1. The experiment consists of independent and identical trials.
2. Each trial results in one of the two outcomes called success or failure.
3. The probability of success in a single trial is p and remains the same from trial to trial. The probability of a failure, also in a single trial is $q = 1 - p$
4. The random variable of interest, X is the number of successes observed during the n trials.

Binomial Distribution

Theorem

Let X be a binomial random variable with parameters n and p. Then $\mathbb{P}$ the probability mass function of X, is

$$\mathbb{P}(X = x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & ; x = 0, 1, 2, \cdots n \\ 0 \quad \textit{otherwise} \end{cases} \tag{7}$$

We can show that (7) is a pmf

If X is a binomial random variable with parameters n and p, (shorthand notation is $X \sim B(n, p)$) then

$$\mathbb{E}(X) = np, \quad \mathbb{V}(X) = np(1-p), \quad \sigma = \sqrt{np(1-p)} \tag{8}$$

We can show proofs of (8)

Example

Four fair coins are flipped. If their outcomes are assumed independent,

1. What is the probability that two heads and two tails are obtained.
2. Calculate the mean and variance
3. Find $\mathbb{P}(0 < X < 4 \mid X > 2)$

Solution

Let X equals the number of heads ("success") that appear then X is a binomial random variable with parameters ($n = 4$ *and* $p = \frac{1}{2}$) and pmf given by,

$$\mathbb{P}(X = x) = \binom{4}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x} \quad x = 0, 1, 2, 3, 4$$

1. $\mathbb{P}(X = x) = \binom{4}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{3}{8}$
2. $\mathbb{E}(X) = np = 4\left(\frac{1}{2}\right) = 2$

$$\mathbb{V}(X) = np(1-p) = 4\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 1$$

Solution Cont'd

$$\mathbb{P}(0 < X < 4 \mid X > 2) = \frac{\mathbb{P}(0 < X < 4 \cap X > 2)}{\mathbb{P}(X > 2)}$$
$$= \frac{\mathbb{P}(X = 3)}{\mathbb{P}(X = 3) + \mathbb{P}(X = 4)} = \frac{0.25}{0.3125} = \frac{4}{5}$$

Example

Question It is known that any item produced by a certain machine will be defective with probability 0.1, independently of any other item. What is the probability that in a sample of three items, at most one will be defective?

Solution

If X is the number of defective items in the sample, then X is a binomial random variable with parameters $(3, 0.1)$. Hence, the desired probability is given by

$$\mathbb{P}\{X = 0\} + \mathbb{P}\{X = 1\} = \binom{3}{0}(0.01)^0(0.9)^3 + \binom{3}{1}(0.01)^1(0.9)^2$$
$$= 0.972$$

Que What is the probability that at least one will be perfect?

Exercises

Exercise A restaurant serves 8 entrées of fish, 12 of beef, and 10 of poultry. If customers select from these entrées randomly, what is the probability that two of the next four customers order fish entrees? Ans $= 0.23$

Poisson Distribution

The Poisson distribution is used as a model for describing the number of times some random event occurs in an interval of time or an area, a volume or other unit.

Some examples are the number of claims processed by a certain insurance company in a given month,

The number of road traffic accidents in an area during a given time interval, the number of errors a typist makes in typing a page of a text and the number of admissions of a clinic in a given time interval.

Poisson

1. the number of customers entering a post office in a given hour
2. the number of α-particles discharged from a radioactive substance in one second
3. the number of machine breakdowns per month
4. the number of defects on a piece of raw material.

Poission Distribution

Definition (Poisson Distribution)

A discrete random variable X with possible values 0, 1, 2, 3, $\cdots$ is called Poisson with parameter λ, $\lambda > 0$, if

$$\mathbb{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!} \quad k = 0, 1, 2, 3 \cdots$$

In general, we define X = the number of "occurrences" over a unit interval of time (or space).

If X is a Poisson random variable with parameter λ written as $X \sim$ Poisson(λ), then

$$\mathbb{E}(X) = \lambda, \quad \mathbb{V}(X) = \lambda, \quad \sigma = \sqrt{\lambda} \tag{9}$$

Example

The number of power surges in an electric grid has a Poisson distribution with a mean of one power surge every twelve hours.
a. What is the probability that there will be no more than one power surge in a 24-hour period?
b. What is the probability that there will be more than three power surge in a 24-hour period?

Solution

In 12 hours, the mean number of power surges is 1. In 24 hours, the mean number is therefore 2.

$$\begin{aligned} P(X \leq 1) &= P(X = 0) + P(X = 1) \\ &= \frac{2^0 e^{-2}}{0!} + \frac{2^1 e^{-2}}{1!} \\ &= e^{-2} + 2e^{-2} \\ &= 3e^{-2} \\ &= 0.406 \end{aligned}$$

Solution Cont'd

$$
\begin{aligned}
P(X > 3) &= 1 - P(X \leq 3) \\
&= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)] \\
&= 1 - \left[\left(\frac{2^0 e^{-2}}{0!}\right) + \left(\frac{2^1 e^{-2}}{1!}\right) + \left(\frac{2^2 e^{-2}}{2!}\right) + \left(\frac{2^3 e^{-2}}{3!}\right)\right] \\
&= e^{-2} + 2e^{-2} + 2e^{-2} + \frac{4e^{-2}}{3} \\
&= 0.857
\end{aligned}
$$

The desired probability is $1 - 0.857 = 0.143$

Poisson Approximation to Binomial

When the number of trials, n in a Binomial process is large, the computations of the binomial probabilities may be too tedious. The Poisson distribution can be used as an alternative to approximate the Binomial distribution.

Exercises

1. Suppose that on average, 1 person in 1000 makes a numerical error in preparing his or her income tax return. If 9000 forms are selected at random and examined, find the probability that less than 3 of the forms contain an error. Ans $= 0.0062$
2. Every week the average number of wrong-number phone calls received by a certain mail-order house is seven. What is the probability that they will receive (a) two wrong calls tomorrow; (b) at least one wrong call tomorrow? Answers (a) ≈ 0.18 (b) ≈ 0.63

Exercise

1. Orders arrive at a Web site according to a Poisson process with a mean of 12 per hour. Determine the following:
   (a) Probability of no orders in five minutes.
   (b) Probability of 3 or more orders in five minutes.
   (c) Length of a time interval such that the probability of no orders in an interval of this length is 0.001.

Continuous Random Variables

- A continuous random variable is a random variable with an interval (either finite or infinite) of real numbers for its range.
- For example, if X = time (measured in seconds), then the set of all possible values of X is

$$\{x : x > 0\}$$

- If X = temperature (measured in degree Celsius), the set of all possible values of X (ignoring absolute zero and physical upper bounds) might be described as

$$\{x : -\infty < x < +\infty\}$$

These set of values cannot be counted

Probability Density Function

In continuous random variables we assign positive probability to events which are intervals ($e.g., 2 < X < 4, etc.$) The probability density function (pdf), denoted by $f(x)$, has the following characteristics:

- $f(x) \geq 0$
- The area under any pdf is equal to 1,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

- If t is a specific value of interest, then the cummulative distribution function (cdf) of X is given by

$$F(t) = \mathbb{P}(X \leq t) = \int_{-\infty}^{t} f(x)dx$$

Probability Density Function

We can view $f(x)$ as the first derivative of $F(x)$, i.e

$$f(x) = F'(x)$$

- If x_1 and x_2 are specific values of interest $(x_1 < x_2)$, then

$$\begin{aligned} \mathbb{P}(x_1 \leq X \leq x_2) &= \int_{x_1}^{x_2} f(x)dx \\ &= F(x_2) - F(x_1) \end{aligned} \tag{10}$$

- For a fixed value t, $\mathbb{P}(X = t) = 0 = \int_t^t f(x)dx = 0$ from equation (10).

Probability Density Function

Consequently

$$\mathbb{P}(x_1 \leq X \leq x_2) = \mathbb{P}(x_1 < X \leq x_2) = \mathbb{P}(x_1 \leq X \leq x_2) = \mathbb{P}(x_1 < X < x_2)$$

with each equal to

$$\int_{x_1}^{x_2} f(x)dx$$

which is not true if X has a discrete distribution.

Example

X has the following density;

$$f(x) = 4x^3 \text{ where } 0 \leq x \leq 1$$

Find
(i) F(x)
(ii) $\mathbb{P}(0.3 \leq X \leq 0.7)$ and $\mathbb{P}(0.4 \leq X \leq 0.6)$

Solution

$$F(x) = \int_{-\infty}^{x} f(t)dt = \int_{0}^{x} 4t^3 dt = t^4\Big|_0^x = x^4$$

$$P(0.3 \leq x \leq 0.7) = F(0.7) - F(0.3) = (0.7)^4 - (0.3)^4 = 0.232$$

$$P(0.4 \leq x < 0.6) = F(0.6) - F(0.4) = (0.6)^4 - (0.4)^4 = 0.104$$

Example

Let X be a continuous random variable with cumulative distribution function given by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{9}x^3 & 0 \leq x \leq 3 \\ 1 & x \geq 3 \end{cases}$$

Find the p.d.f of X

Solution

$$f(x) = \frac{dF(x)}{dx} = \begin{cases} \frac{1}{3}x^2 & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Expectation and Variance

The expected value of X is given by

$$\mu = \mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x)dx$$

And the variance of X is given as

$$\sigma^2 = \mathbb{V}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2$$

The standard deviation of X given as

$$\sigma = \sqrt{\sigma^2} = \sqrt{\mathbb{V}(X)}$$

Expectation and Variance

Let X be a continuous random variable with pdf $f(x)$. Suppose that $g, g_1, g_2, \cdots, g_k$ are real-valued functions, and let c be any real constant. then

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Expectation satisfy the following properties

- $\mathbb{E}(c) = c$
- $\mathbb{E}(cg(X)) = c\mathbb{E}(g(X))$

Given a linear function $g(x) = ax + b$ where a, b are constants, we have

$$\mathbb{E}(g(X)) = a\mathbb{E}(X) + b \text{ and } \mathbb{V}(aX + b) = a^2\mathbb{V}(X)$$

Example

Suppose that X has the pdf

$$f(x) = \begin{cases} 3x^2 \ ; 0 < x < 1 \\ 0 \ \text{otherwise} \end{cases} \tag{11}$$

Find the

- Find the cdf of X
- Calculate $\mathbb{P}(X < 0.3)$
- Calculate $\mathbb{P}(X > 0.8)$
- Calc. $\mathbb{P}(0.3 < X < 0.8)$
- Find $\mathbb{E}(X)$
- Find the standard deviation of X
- If we define $Y = 3X$, find the cdf and pdf of Y. Further calculate the mean and variance of Y .

Example

Suppose X has the following pdf, where A is a constant to be determined.

$$f(x) = \begin{cases} A(1-x^2) & ; -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find A, $\mathbb{P}\{0.5 < X < 1.5\}$, CDF (F_X), $\mathbb{E}(X)$, $\mathbb{V}(X)$, and σ_X

Solution

To find A

$$\int_{-\infty}^{\infty} f(u)du = 1$$

$$= \int_{-1}^{1} A(1-u^2)du = A\left(u - \frac{u^3}{3}\right)\Big|_{-1}^{1} = 1$$

Hence $A = \frac{3}{4}$

$$\mathbb{P}(0.5 < X < 1.5) = \int_{0.5}^{1.5} f(u)du = \int_{0.5}^{1} \frac{3(1-u^2)}{4}du = \frac{3}{4}\left(u - \frac{u^3}{3}\right)\Big|_{0.5}^{1}$$

$$= \frac{3}{4}\left(\frac{2}{3} - \frac{11}{24}\right) = \frac{5}{32}$$

Solution

$$F(x) = \begin{cases} 0 & c < -1 \\ ? & -1 < c < 1 \\ 1 & c \geq 1 \end{cases}$$

Let us determine the "?", that is the value of $F(c)$ for $-1 \leq c < 1$

$$\begin{aligned} F(c) &= \mathbb{P}(X \leq c) \\ &= \int_{-1}^{c} \frac{3(1-u^2)}{4} du = \frac{3}{4}\left(u - \frac{u^3}{3}\right)\Big|_{-1}^{c} = \frac{2+3c-c^3}{4} \end{aligned}$$

Solution

$$F(x) = \begin{cases} 0 & c < 1 \\ \frac{2+3c-c^2}{4} & -1 < c < 1 \\ 1 & c \geq 1 \end{cases}$$

Find $\mathbb{E}(X) = \int_{-1}^{1} \frac{3}{4} u(1-u^2) du = 0$,

$$\mathbb{V}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \mathbb{E}(X^2)$$

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} u^2 f(u) du = \int_{-1}^{1} u^2 \frac{3}{4}(1-u^2) du$$

$$= \frac{3}{4} \int_{-1}^{1} (u^2 - u^4) du = 0.2$$

Thus $\mathbb{V}(X) = 0.2$ and $\sigma_X = \sqrt{0.2} = 0.45$

Exercise

Experience has shown that while walking in a certain park, the time X, in minutes, between seeing two people smoking has a density function of the form

$$f(x) = \begin{cases} \lambda x e^{-x} & ; x > 0 \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

- Calculate the value of λ, **Ans**, $\lambda = 1$
- Find the probability distribution function of X
- What is the probability that Jeff, who has just seen a person smoking, will
  (i) see another person smoking in 2 to 5 minutes?.
  Ans ≈ 0.37
  (ii) In at least 7 minutes? **Ans** $= 0.007$

Exercise

When a certain car breaks down, the time that it takes to fix it (in hours) is a random variable with the density function

$$f(x) = \begin{cases} ce^{-3x} & ; 0 \leq x < \infty \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

1. Calculate the value of c
2. Find the probability that when this car breaks down, it takes at most 30 minutes to fix it.

Exercise

Let X be a random variable with pdf

$$f(x) = \begin{cases} 3x^2 \ ; 0 < x < 1 \\ 0 \ \text{otherwise} \end{cases}$$

(i) Find $\mathbb{E}(2X + 4)$

Exponential Distribution

The Exponential Distribution is commonly used to answer the following questions:

- How long do we need to wait before a customer enters a shop?
- How long will it take before a call center receives the next phone call?
- How long will a piece of machinery work without breaking down?
- All these questions concern the time we need to wait before a given event occurs. We often model this waiting time by assuming it follows an exponential distribution.

Exponential Distribution

A random variable X is said to have an exponential distribution with parameter $\lambda > 0$ if its pdf is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & ; x > 0 \\ 0 & \text{otherwise} \end{cases}$$

The shorthand notation is $X \sim Exp(\lambda)$. The CDF of X is

Exponential Distribution

For any specified time x, the probability of the event happens no later than x is

$$F(x) = \int_{-\infty}^{x} f(t)dt = \int_{0}^{x} \lambda e^{-\lambda t} dt \ = \begin{cases} 1 - e^{-\lambda x} & x > 0 \\ 0 & x < 0 \end{cases}$$

The expected value of X is

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x) = 0 + \int_{0}^{\infty} x\lambda e^{-\lambda x} \tag{14}$$

Integrating by parts, setting $u = x$ and $dv = \lambda e^{-\lambda x} dx$ we obtain

$$\mathbb{E}(X) = \frac{1}{\lambda}$$

Exponential Distribution

and the variance of X given

$$\mathbb{V}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

For an exponential random variable with parameter λ,

$$\mathbb{E}(X) = \frac{1}{\lambda} = \sigma_X, \quad \mathbb{V}(X) = \frac{1}{\lambda^2}$$

Example

Suppose X has the exponential distribution with mean 10.
Determine the following
(i) $\mathbb{P}(X > 10)$
(ii) $\mathbb{P}(X > 30)$
(iii) that value of x such that $\mathbb{P}(X < x) = 0.95$

Solution

$$f(x) = \frac{1}{10} e^{-x/10}, \qquad x \geq 0$$

$$\begin{aligned} P(x > 10) &= \int_{10}^{\infty} f(t)dt = \frac{1}{10} \int_{10}^{\infty} e^{-t/10} dt \\ &= -e^{-t/10} \Big|_{10}^{\infty} = e^{-1} = 0.3678 \\ P(X > 30) &= 1 - P(X \leq 30) = 1 - F(30) = 1 - (1 - e^{-30/10}) \\ &= e^{-3} = 0.0498 \end{aligned}$$

Solution

$$
\begin{aligned}
P(X < x) &= 0.95 \\
F(x) &= 0.95 \\
1 - e^{-x/10} &= 0.95 \\
e^{-x/10} &= 0.05 \\
x &= -10 \ln 0.05 \\
x &= 29.96
\end{aligned}
$$

Example

Assume that the length of a phone call in minutes is an exponential random variable X with parameter $\lambda = \frac{1}{10}$, (the question may tell you the value of λ through the expectation; i.e., this based the expected waiting time for a phone call is 10 minutes). If someone arrives at a phone booth just before you arrive,

find the probability that you will have to wait
(a) less than 5 minutes
(b) greater than 10 minutes
(c) between 5 and 10 minutes
Also compute the expected value and variance.

Uniform Distribution

The uniform distribution provides a simple probability model to describe a continuous random variable that can randomly assume any value between two points a and b ($a < b$) on a line. It therefore provides a good model for a continuous random variable whose values are uniformly distributed over an interval.

Uniform Distribution

Let X be a uniform random variable on the interval (a,b). The corresponding pdf denoted by $f(x)$ is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

Show that equation (15) is a density function.

Expectation and Variance

The expected value of X is

$$\mathbb{E}(X) = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \frac{x^2}{2}\Big|_a^b = \frac{1}{b-a}\left(\frac{1}{2}b^2 - \frac{1}{2}a^2\right) = \frac{a+b}{2}$$

The variance is given as

$$\mathbb{E}(X^2) = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{3}\frac{b^3-a^3}{b-a} = \frac{1}{3}(a^2+ab+b^2)$$

$$\mathbb{V}(X) = \frac{(b-a)^2}{12}$$

Uniform Distribution

Its CDF is given as

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & x \geq b \end{cases} \tag{16}$$

We will that equation (16) is the CDF of a uniform random variable.

Example

If X is uniformly distributed over $(0, 10)$, calculate the probability that
(a)$X < 3$ (b) $X > 6$, and (c) $3 < X < 8$.

$$\mathbb{P}(X < 3) = \int_0^3 \frac{1}{10} dx = \frac{3}{10}$$

$$\mathbb{P}(X > 6) = \int_6^{10} \frac{1}{10} dx = \frac{4}{10}$$

$$\mathbb{P}(3 < X < 8) = \int_3^8 \frac{1}{10} dx = \frac{1}{2}$$

Example

A bus arrives every 20 minutes at a bus stop. It is assumed that the waiting time for a particular individual is a random variable with continuous uniform distribution.

(a) Compute the mean and standard deviation of an individual's waiting time.

(b) Find the probability that an individual waits more than 9 minutes.

(c) Find the probability that an individual waits between 2 and 10 minutes.

Solution

The probability distribution for the waiting time, X

$$f(x) = \begin{cases} \frac{1}{20} & 0 \leq x \leq 20 \\ 0 & \text{elsewhere} \end{cases}$$

The mean and standard deviation of X are

$$\begin{aligned} E(X) &= \frac{b+a}{2} = \frac{0+20}{2} = 10 \\ Var(X) &= \frac{(b-a)^2}{12} = \frac{(20-0)^2}{12} = 33.333 \end{aligned}$$

Solution

$$\sigma = \sqrt{33.33333} = 5.7735$$

$$P(X > 9) = \int_9^{20} \frac{1}{20} dx = \frac{x}{20}\Big|_9^{20} = \frac{20}{20} - \frac{9}{20} = \frac{11}{20}$$

$$P(2 < X < 10) = \int_2^{10} \frac{1}{20} dx = \frac{x}{20}\Big|_2^{10} = \frac{10}{20} - \frac{2}{20} = \frac{8}{20}$$

Example

Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits
(a) less than 5 minutes for a bus
(b) more than 10 minutes for a bus.

Solution

Let X denote the number of minutes past 7 that the passenger arrives at the stop. Since X is a uniform random variable over the interval (0, 30), it follows that the passenger will have to wait less than 5 minutes if (and only if) he arrives between 7:10 and 7:15 or between 7:25 and 7:30. Hence, the desired probability for part (a) is

Solution

$$\mathbb{P}(10 < X < 15) + \mathbb{P}(25 < X < 30) = \int_{10}^{15} \frac{1}{30} dx + \int_{25}^{30} \frac{1}{30} dx = \frac{1}{3}$$

(b) He would have to wait more than 10 minutes if he arrives between 7 and 7:05 or between 7:15 and 7:20, so the probability for (b) is

$$\mathbb{P}(0 < X < 5) + \mathbb{P}(15 < X < 20) = \frac{1}{3}$$

Normal Distribution

X is a normal random variable, or simply that X is normally distributed, with parameters μ and σ^2 if the density of X is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad -\infty < x < \infty \tag{17}$$

Shorthand notation is $X \sim N(\mu, \sigma)$. Another name for the normal distribution is the Gaussian distribution. We state without proofs;

$$\mathbb{E}(X) = \mu \;\; \mathbb{V}(X) = \sigma^2$$

Example: If $X \sim N(1, 4)$,

find the mean, variance, and standard deviation of X.

Normal Distribution

Properties of the Normal Distribution

Definition: A **normal distribution** is a continuous, symmetric, bell-shaped distribution of a variable.

- A normal distribution curve is bell-shaped.
- The mean, median and mode are equal and are located at the centre of the distribution.
- A normal distribution curve is unimodal (i.e., it has only one mode).
- The curve is symmetric about the mean.
- The curve is continuous; but it never touches the x-axis.
- The total area under a normal distribution curve is equal to 1.00, or 100%.
- The area under the part of a normal curve that lies within 1 standard deviation of the mean is approximately 0.68, or 68%; within 2 standard deviations, about 0.95, or 95%; and within 3 standard deviations, about 0.997, or 99.7%.

Normal Distribution

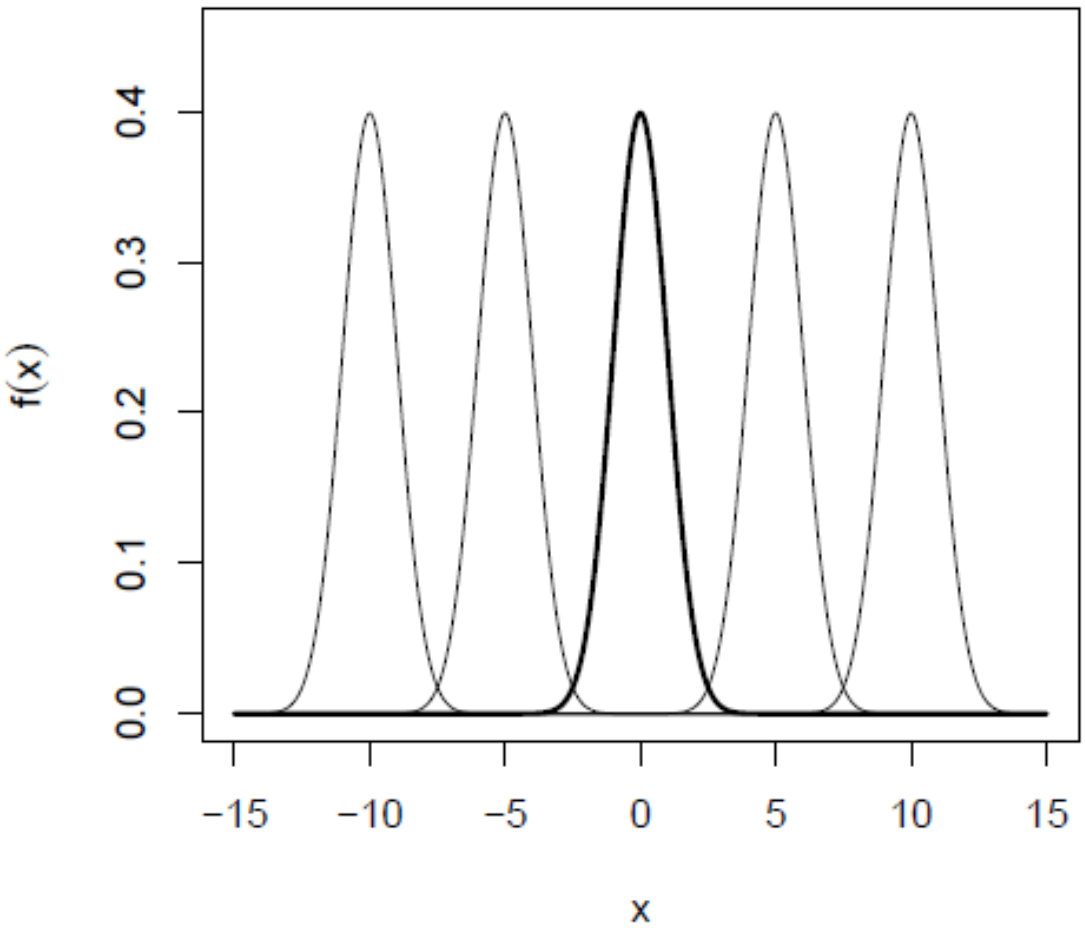


Figure 1: Normal Distribution Plots

Figure (1) presents the plot of pdf of N(-10, 1), N(-5, 1), N(0, 1),

Normal Distribution

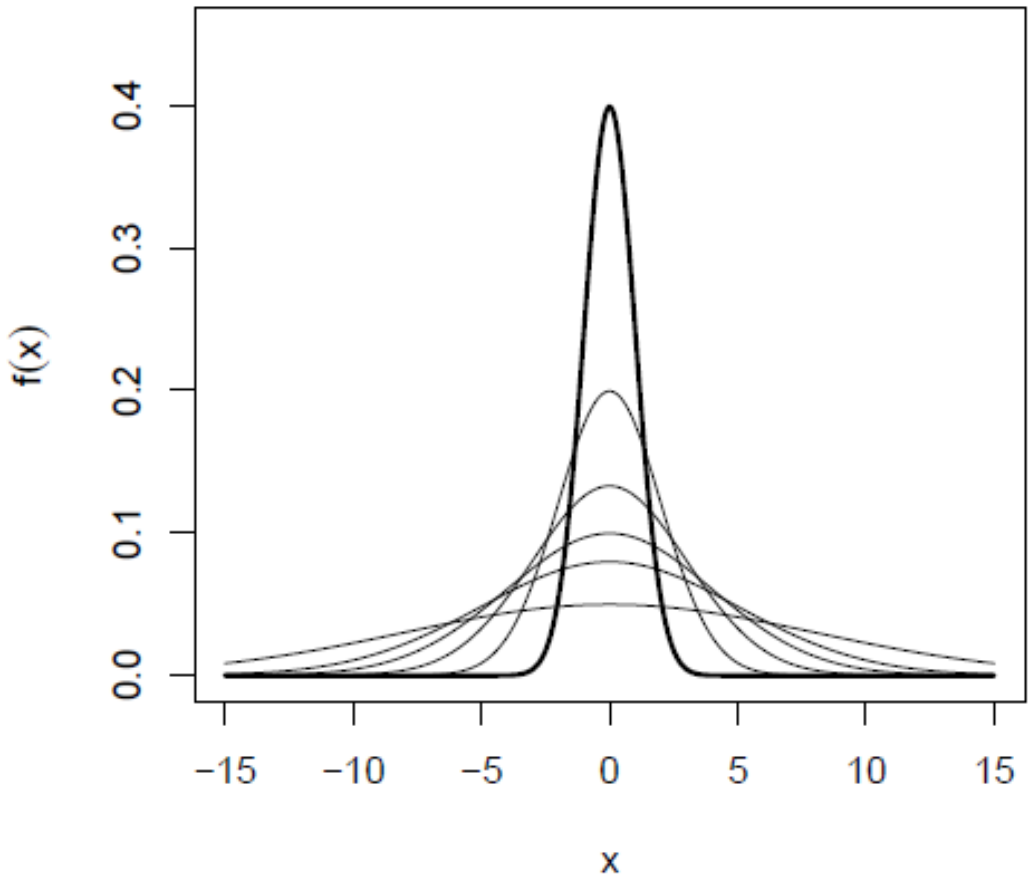
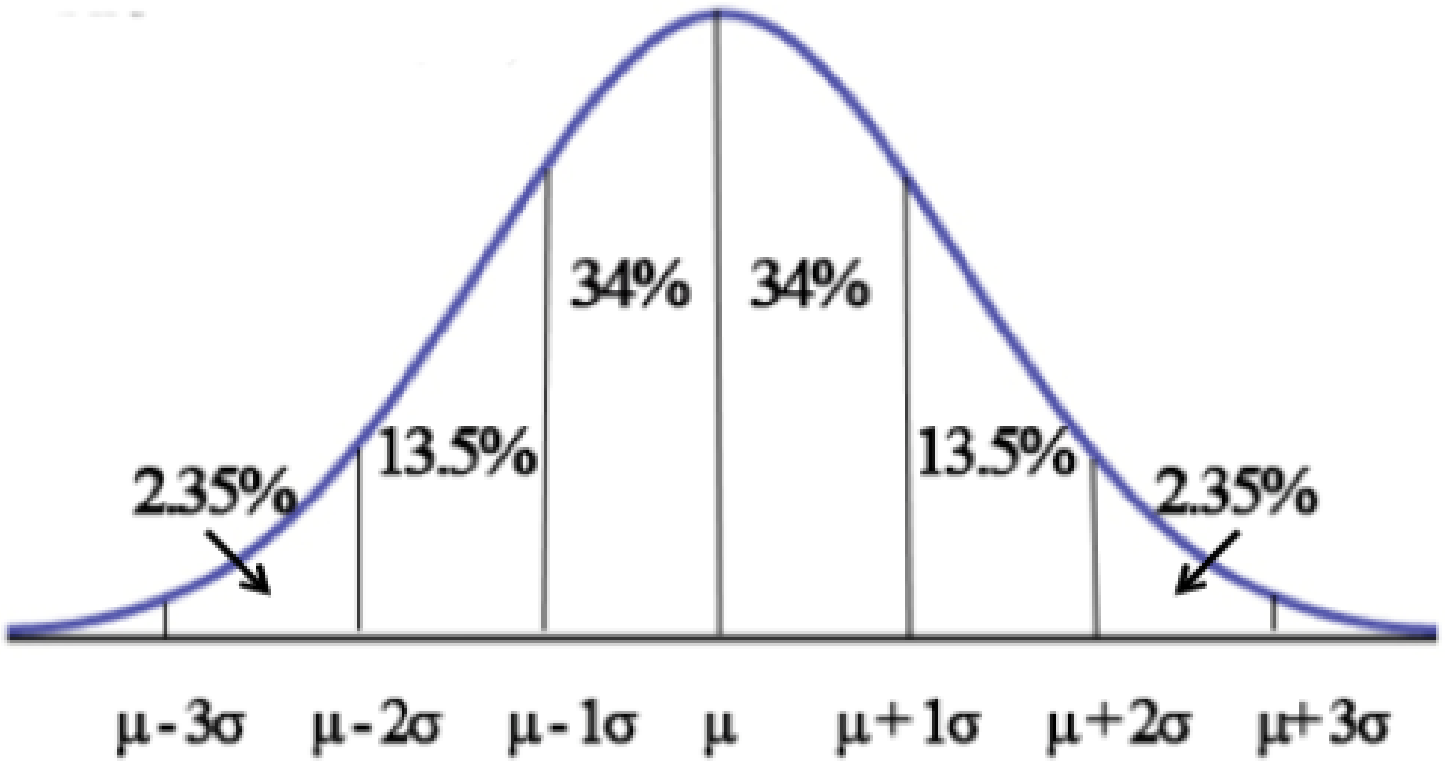


Figure 2: Normal Distribution Plots

Figure (2) presents the plot of pdf of N(0, 1), N(0, 4), N(0,9),

Empirical Rule



Example

The marks obtained by 600 engineering students in a test are normally distributed with a mean of 75 and a standard deviation of 7.
(a) What is the proportion of the class that has a test score between 68 and 82
(b) How many students have a test score between 61 and 89
(c) What is the probability that a student chosen at random has a test between 54 and 75.

Solution

Using the empirical rule

$$(a)\ \ \mathbb{P}(68 < X < 82) = 0.68$$

$$(b)\ \ \mathbb{P}(61 < X < 89) = 0.95$$

Hence the number of students is $600 \times 0.95 = 570$ students

$$(c)\ \ \mathbb{P}(54 < X < 75) = 0.4985$$

Solution

We can also use equation (17) to compute the probabilities

$$(a)\ \mathbb{P}(68 < X < 82) = \frac{1}{7\sqrt{2\pi}} \int_{68}^{82} e^{-\frac{(x-75)^2}{2(7^2)}} dx = 0.68268949$$

$$(b)\ \mathbb{P}(61 < X < 89) = \frac{1}{7\sqrt{2\pi}} \int_{61}^{89} e^{-\frac{(x-75)^2}{2(7^2)}} dx = 0.9544997$$

$$(c)\ \mathbb{P}(54 < X < 75) = \frac{1}{7\sqrt{2\pi}} \int_{54}^{75} e^{-\frac{(x-75)^2}{2(7^2)}} dx = 0.498650102$$

Standard Normal Distribution

When $\mu = 0$ and $\sigma^2 = 1$, we say the normal distribution $N(0, 1)$ is the standard normal distribution or the z score for X. It is denoted by Z, i.e

$$Z \sim N(0, 1)$$

For $X \sim N(\mu, \sigma)$

$$\frac{X - \mu}{\sigma} = Z \sim N(0, 1)$$

The transformation process is referred to as standardization. In other words we can express

$$X = \mu + \sigma Z$$

Thus, all the normal distribution shares a common thing which is the standard normal distribution. If we know standard normal, we know every normal distribution.

Standard Normal Distribution

Example 1 If $X \sim N(3, 16)$ then the corresponding z score for $x = 8$ after standardization is

$$z = \frac{8 - 3}{4} = 1.25$$

Example 2 Given that $X \sim N(30, 10)$. Find the mean and standard deviation.
(b) What is the value of x with the z - score of 1.5
(c) what is the z -score that correspond to $x = 30$

Standard Normal Distribution

The cumulative distribution function of a standard normal random variable is denoted by $\Phi(x)$ and is given as

$$\Phi(x) = \mathbb{P}(Z \leq x) = \mathbb{P}(X \leq \sigma x + \mu) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\sigma x+\mu} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

Letting $y = \frac{t-\mu}{\sigma}$ we obtain

$$\mathbb{P}(Z \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-y^2/2} dy$$

Standard Normal Distribution

Its pdf is given as

$$\Phi'(x) = f(x) = \frac{1}{2\sqrt{\pi}} e^{-x^2/2} \tag{18}$$

For negative values of x, $\Phi(x)$ can be obtained from the relationship

$$\mathbb{P}(Z > x) = \Phi(-x) = 1 - \Phi(x)$$

Finding Probabilities of Random Normal Variables

The standard normal table and our ability to standardize any normal random variable allow us to determine the probability of normal random variable.

Find $\mathbb{P}(3 < X < 8)$. Where $X \sim N(3, 16)$
This is equivalent to

$$\mathbb{P}\left(\frac{3-3}{4} < \frac{x-\mu}{\sigma} < \frac{8-3}{4}\right) = \mathbb{P}(0 < Z < 1.25) = 0.3944$$

We read from the standard normal tables.

Example

Find $\mathbb{P}(-1.96 < Z < 1.96)$

$$\begin{aligned}\mathbb{P}(-1.96 < Z < 1.96) &= \mathbb{P}(-1.96 < Z < 0) + \mathbb{P}(0 < Z < 1.96) \\ &= 0.9500\end{aligned}$$

Suppose a continuous random variable X is normally distributed with mean 30 and standard deviation 5.

Find
(ii) $\mathbb{P}(20 < X < 30)$ Ans $= 0.4772$
(iii) $\mathbb{P}(30 < X < 37.5)$ Ans $= 0.4332$

Example

X is normally distributed with mean 100 and standard deviation 15. Use the standard z table to answer the following questions. (i) $\mathbb{P}(X < 80)$ Ans $= 0.09$
(ii) $\mathbb{P}(X > 136)$ Ans $= 0.0082$
(iii) $\mathbb{P}(95 < X < 110)$ Ans $= 0.3779$
(iv) The X score that corresponds to the 90th percentile (z =1.285 and X =119.2)

Standard Normal Table

Finding Areas Under the Standard Normal Distribution Curve

1. Find the area to the left of $z = 1.59$
2. Find $P(z < -1.42)$
3. Find the area to the right of $z = 2.09$
4. Find $P(z > 0.09)$
5. Find the area between $z = -1.96$ and $z = 2.58$
6. Find $P(-1.25 < z < 2.50)$

Solution

1. Cumulative area under the standard normal distribution curve for $z = 1.59$ is 0.9441.
2. $P(z < -1.42) = 0.0778$
3. Cumulative area for $z = 2.09$ is 0.9817; thus the required area is $1 - 0.9817 = 0.0183$
4. $P(z > 0.09) = 1 - 0.5359 = 0.4641$
5. Cumulative area for $z = -1.96$ is 0.0250 and cumulative area for $z = 2.58$ is 0.9951. Thus, the required region is $0.9951 - 0.0250 = 0.9701$
6. $P(-1.25 < z < 2.50) = 0.9938 - 0.1056 = 0.8882$

Examples

Finding z scores

Eg. 1 Find the z score corresponding to the 95th percentile.

Solution

From the z table, 0.9500 lies exactly between 1.64 and 1.65 and thus, $z = 1.645$.

Eg. 2 Find z such that 60% of the area under the curve falls to its right.

Solution

The cumulative area is 40%, i.e. 0.4000 whose closest table value is 0.4013 corresponding to $z = -0.25$.

Eg. 3 Find the value of z such that 80% of the area under the curve lies between $-z$ and z.

Solution

$-z$ corresponds to a cumulative area of 10%, i.e. 0.1000. The table value is 0.1003 which corresponds to -1.28. Since the normal distribution is symmetrical, it implies 80% of the area would lie between -1.28 and 1.28.

Thus, $z = 1.28$

Example

Applications

Eg.1 A survey by the National Retail Federation found that women spend on average \$146.21 for the Christmas holidays. Assume the standard deviation is \$29.44. Find the percentage of women who spend less than \$160.00. Assume the variable is normally distributed.

Solution

$$X = \$160.00,\ \mu = \$146.21\ and\ \sigma = \$29.44\ ;\ P(X < \$160) = ?$$

$$z = \frac{X-\mu}{\sigma} = \frac{160-146.21}{29.44} = 0.47$$

$$P(z < 0.47) = 0.6808$$

Thus, 68.08% of women spend less than $\$160$.

Example

Eg. 2 Each month, an American household generates an average of 28 pounds of newspaper for garbage or recycling. Assume the standard deviation is 2 pounds. If a household is selected at random, find the probability of its generating

a) Between 27 and 31 pounds per month

b) More than 30.2 pounds per month

Assume the variable is approximately normally distributed.

Solution

$$\mu = 28\ lbs \quad and \quad \sigma = 2\ lbs$$

a) $P(27 < X < 31) = ?$ $\quad z = \frac{27-28}{2} = -0.50\ ;$ $\quad z = \frac{31-28}{2} = 1.50$

$$\Rightarrow P(-0.50 < z < 1.50) = 0.9332 - 0.3085 = 0.6247$$

b) $P(X > 30.2) = ?$ $\quad z = \frac{30.2-28}{2} = 1.10$

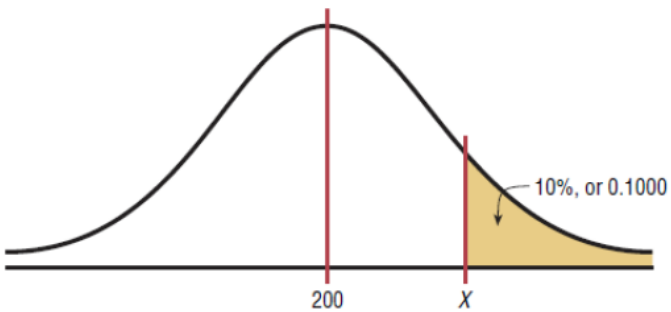
$$\Rightarrow P(z > 1.10) = 1 - 0.8643 = 0.1357$$

Example

Eg. 3 To qualify for a police academy, candidates must score in the top 10% on a general abilities test. The test has a mean of 200 and a standard deviation of 20. Find the lowest possible score to qualify. Assume the test scores are normally distributed.

Solution

Since the test scores are normally distributed, the test value X that cuts off the upper 10% of the area under a normal distribution curve is desired.



The cumulative area is $1 - 0.1000 = 0.9000$. The closest table value is 0.8997 which corresponds to $z = 1.28$.

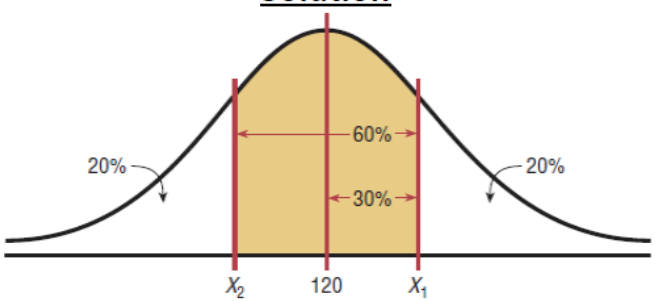
Thus, $z = \frac{X-\mu}{\sigma} \implies 1.28 = \frac{X-200}{20}$; $20(1.28) = X - 200$; $X = 200 + 20(1.28) = 226$

A score of 226 should be used as a cut-off. Anybody scoring 226 or higher qualifies.

Example

Eg. 4 For a medical study, a researcher wishes to select people in the middle 60% of the population based on blood pressure. If the mean systolic blood pressure is 120 and the standard deviation is 8, find the upper and lower readings that would qualify people to participate in the study. Assume that blood pressure readings are normally distributed.

Solution



To get the area to the left of the positive z value, add $0.5000 + 0.3000 = 0.8000$ with the closest table value being 0.7995; which corresponds to $z = 0.84$.

$X_1 = \mu + z\sigma = 120 + 0.84(8) = 126.72$

The area to the left of the negative z value is 20%, or 2.000; with table value -0.84 .

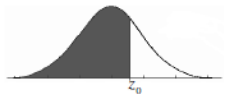
$X_2 = 120 - 0.84(8) = 113.28$

Therefore, the middle 60% will have blood pressure readings of $113.28 < \mathrm{X} < 126.72$.

Table

Table 2 Area under the Standard Normal Distribution to the Left of z_0: Positive z_0

$$\Phi(z_0) = P(Z \le z_0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z_0} e^{-x^2/2}\, dx$$



z_0	0	1	2	3	4	5	6	7	8	9
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7703	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9889	.9889	.9889	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998
3.6	.9998	.9998	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999
3.7	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999
3.8	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999

Note that for $z_0 > 3.89$, $\Phi(z_0) = P(Z \le z_0) \approx 1$.

Extra Reading

Read on:

1. mean, median, mode (measures of central tendency)
2. variance, standard deviation, range (measures of variation)